Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2005
PI Name:	Roubenoff, Ronenn M.D.		
Project Title:	Timed Feeding and Resistance Training to Prevent Muscle Atrophy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02111	**Congressional District:**	1
Comments:	First name changed from Ronnen to Ronenn per Tufts University website -- rje 2/1/2017		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	09/01/2003	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2004 with PI change to Castaneda-Sceppa at that time [ed. Nov 2010]		
Key Personnel Changes/Previous PI:	PI changed to Castaneda-Sceppa in 2004; see Castaneda-Sceppa, Carmen, for reports in FY2005 and later (note added 2/2009).		
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NPFR00301		
Performance Goal No.:			
Performance Goal Text:			

The plan for year two is to actively continue subject enrollment into the study. We plan to enroll 13 more subjects and have preliminary data available for analyses. We do not plan to change the original hypothesis, aims or research plan in any way. The aims of this study will remain as stated under the specific aims section above.

The logistical difficulties of studying alterations in muscle during or even after space missions have stimulated the development of ground-based models, such as bed rest, to simulate weightlessness. In general, reductions in muscle strength and cross-sectional area during bed rest are very similar to those seen with space flight. For example, thirty days of bed rest produce an 18-20% decrease in isokinetic strength of the knee extensors, a 6% decrease in the knee flexors, and an 8% reduction in mid-thigh muscle CSA. Additionally, it has been shown that knee extensor strength, whole

Task Description:	muscle cross-sectional area, and vastus lateralis fiber cross-sectional area decreases by 25-30% (various angular velocities), 14%, and 18%, respectively, after 6 wk bed rest. An important question relevant to space flight conditions, as well as to many clinical situations resulting in prolonged bed rest, is the extent of loss during immobilization in addition to the length of recovery required after a prolonged stay in space (or after bed rest post-surgery). After prolonged bed rest, significant deficits in muscle strength and muscle cross-sectional area may persist for weeks, suggesting that reambulation is not enough to recover function in a timely manner. The use of concentric and eccentric actions has resulted in strength preservation or enhancement, maintenance of muscle protein synthesis, and prevention of fiber atrophy. The proposed study is the first that we know of to propose to enhance this effect using timed feedings. An understanding of the recovery process can help us design better rehabilitation strategies since recovery in the absence of rehabilitation is incomplete and muscle function is compromised even weeks post-intervention The earth based applicability of this project is to develop an integrated nutrition and exercise countermeasure against these deleterious effects of immobilization. However, the potential of exercise and nutritional interventions to minimize the loss of muscle function and exercise capacity, and the optimal combination of these modalities, are not known. Resistance training is one modality that offers the hope of mitigating or reversing muscle loss induced by weightlessness. Recent studies suggest that the timing of feedings around a bout of resistance training may augment the effect of exercise on muscle protein balance. This is the very question we are investigating.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The main impact of this study is to provide a better understanding of the potential of exercise and nutritional interventions to minimize the loss of muscle mass, muscle function and exercise capacity during prolonged bed rest. Additionally, the findings associated with the possible synergistic effect of timed feeding of an amino acid supplement and resistance exercise on preventing bed rest induced muscle atrophy will be of great impact to the scientific community.
Task Progress:	-Development of an effective resistance exercise intervention accomplished via a pilot study of 5 volunteers who were asked to exercise 3 times a week for 4 weeks. Our exercise routine was refined so that we would anticipate the hypothesized changes in muscle strength. - Staff recruitment and training - Strategies for effective recruitment and retention of study subjects - Bi-weekly study team meetings have allowed for better management and quality control - Monthly conference calls of study investigators with team leader, Dr. Joanne Lupton
Bibliography Type:	Description: (Last Updated:)